

CHAPTER : 2

★ "POINTERS & ARRAYS" ★

POINTERS:-

Definition:-

"Pointers are special variable that have the capability to store memory address of another variable."

This definition is something that you already knew from "Programming Fundamentals", but in this chapter will learn how pointers are at the base of every data structure.

POINTERS IN DS:-

Since data structures is all about arranging the data in memory, pointers are essential to implement data structures. Many data structures require the use of "Dynamic Memory" or "Heap". And you can only access heap using pointers.

Pointers are used in arrays, vectors, linked lists, trees and much more. You might have heard the phrase:

"Array name itself is a pointer."

Yes, the most basic and beginner level data structure is a pointer itself. So in short, pointers are the foundation of data structures. Now let's look at arrays.

ARRAYS:-

Definition:-

"Array is a group of consecutive memory locations reserved under a single identifier, it holds the data of single datatype."

The data of array can be accessed and manipulated using array indexing or pointer arithmetic. Array name is basically a pointer and it contains the base

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address of the array. Now array basically works on the principle of pointer arithmetic. So why do :-

`arr[i];`

What compiler does is :-

`*(arr+i);`

It adds the value to i to the base address to reach the desired element. This also explains why array indexing starts from zero and not one. In order to access the first element, you need to stay on the base address, but if you do :-

`arr[1];`

And compiler does :-

`*(arr+1);`

You would jump to the second element. So to stay on the first element, you need to add zero in base address.

`arr[0];`

And compiler does :-

`*(arr+0);`

`*(arr);`

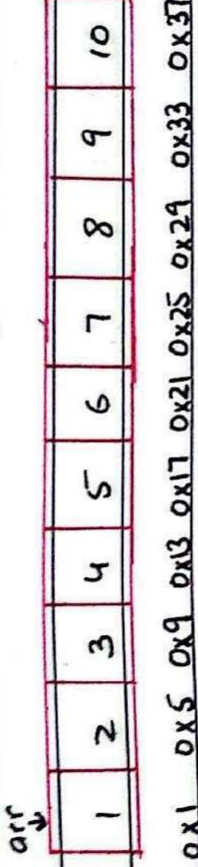
This is how you don't skip the first element.

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MEMORY DIAGRAM :-

Consider : `int arr[10];`



The above diagram shows how an array is structured in the memory. This was all for pointers and arrays, we are not going to go into multidimensional arrays as you have already studied it.

Next up we will study vectors. Normal arrays are static and have a fixed size that cannot be changed on runtime. Vectors cover this drawback of arrays by resizing on runtime. Till Then Good Luck!